

Multi-Electron Capture Enhancement via Heavy Neutral Elements: Scaling Proton–Electron Capture Reactions through High Atomic Number Target Media

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Abstract

We propose that neutral (non-ionized) heavy elements with high atomic number Z provide superior target media for the PECCNAL proton–electron capture mechanism. A neutral atom with Z electrons presents Z sequential capture targets to incoming protons, with each electron remaining available until captured. Unlike hydrogen ($Z = 1$, single capture), heavy neutral atoms such as iron ($Z = 26$), lead ($Z = 82$), or bismuth ($Z = 83$) can sustain up to Z sequential $p + e^- \rightarrow n$ reactions before complete ionization. The reaction compactness and high electron density of heavy elements dramatically increases neutron production per unit volume.

Keywords: heavy elements, multi-electron capture, high- Z target, neutron production, atomic number scaling

1 Introduction

In the original PECCNAL framework [?], hydrogen ($Z = 1$) serves as the reaction medium with one electron per atom. Here we propose that neutral heavy elements offer Z times more capture targets per atom, dramatically increasing volumetric neutron production.

The key requirement is that the element remain **neutral** (non-ionized), ensuring each electron is electrostatically attracted to incoming protons.

2 Multi-Electron Capture Sequence

For a neutral atom with atomic number Z :

$$p + e_{(1)}^- \rightarrow n_1 + \nu_e \tag{H1}$$

$$p + e_{(2)}^- \rightarrow n_2 + \nu_e \tag{H2}$$

$$\vdots$$

$$p + e_{(Z)}^- \rightarrow n_Z + \nu_e \tag{HZ}$$

Each step reduces the atom’s electron count by 1, with the atom progressing from neutral to fully ionized (Z^+) after Z captures.

3 Reaction Rate Scaling

Table 1: Neutron production capacity by target element

Element	Z	Max n/atom	Cost	Availability
Hydrogen (H)	1	1	Low	Abundant
Iron (Fe)	26	26	Low	Abundant
Lead (Pb)	82	82	Low	Abundant
Bismuth (Bi)	83	83	Low	Moderate
Uranium (U)	92	92*	High	Restricted

* Uranium also undergoes fission upon neutron capture: $n + {}^{235}\text{U} \rightarrow 200 \text{ MeV}$

4 Volumetric Neutron Production

For iron ($Z = 26$) vs hydrogen ($Z = 1$) at the same atomic density:

$$\frac{\dot{n}_{\text{Fe}}}{\dot{n}_{\text{H}}} = Z_{\text{Fe}} \times \frac{n_{\text{Fe}}}{n_{\text{H}}} = 26 \times \frac{\rho_{\text{Fe}}/M_{\text{Fe}}}{\rho_{\text{H}}/M_{\text{H}}} = 26 \times \frac{7874/55.85}{0.09/1.008} \approx 26 \times 1590 \approx 41,000 \times \quad (1)$$

Iron provides approximately 41,000 times more neutrons per unit volume than hydrogen gas at STP.

5 Coulomb Barrier Analysis

For neutral atoms, the incoming proton experiences attraction from the outer electron cloud before reaching the nucleus:

$$U(r) = -\frac{e^2}{4\pi\epsilon_0 r} + \frac{Ze^2}{4\pi\epsilon_0 r} \quad (\text{simplified, outer electron dominates}) \quad (2)$$

At large distances ($r > r_{\text{atom}}$), the atom appears neutral and the proton experiences no net force. As it enters the electron cloud, it is attracted to the nearest electron. **No Coulomb barrier exists for proton–electron interaction.**

6 Practical Considerations

Lead (Pb, $Z = 82$) is proposed as optimal:

- Low cost, widely available
- Already used as radiation shielding
- High electron density: $n_e = 3.3 \times 10^{30} \text{ m}^{-3}$

- Stable isotopes, no fission risk
- Liquid at moderate temperatures (melting point 327°C)

7 Conclusion

Neutral heavy elements with high atomic number Z multiply neutron production capacity by up to $Z \times 10^3$ compared to hydrogen gas. Lead ($Z = 82$) is proposed as the optimal practical target, providing high electron density without fission risk. This scaling principle enables compact, high-output PECCNAL reactor designs.

References

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